Sandeep Aryal

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Professional Summary:

Full Stack Software Engineer with over six years of experience designing and developing enterprise applications using Java, Python, and modern cloud technologies. Experienced in building microservices, AI-enabled backends, and RESTful APIs using Spring Boot, Flask, FastAPI, and Django. Skilled in integrating machine learning models into production Java environments and automating data processing with Python. Proficient in AWS services such as ECS, Lambda, and RDS, as well as containerization and orchestration tools including Docker and Kubernetes. Strong foundation in C++, SQL, and database design, with a focus on creating secure and maintainable backend systems. Experienced in mentoring junior developers, conducting code reviews, and guiding best practices across Agile teams to deliver stable, high-performing software solutions that align with business goals.

Core Competencies

Programming Languages: Java, Python, JavaScript, SQL, Bash
Frameworks: Spring Boot, Flask, FastAPI, Django, React.js, Node.js, Angular.js
Microservices & APIs: RESTful APIs, SOAP, GraphQL, Kafka, RabbitMQ, AWS SNS/SQS
Databases: PostgreSQL, AWS RDS, MySQL, MongoDB, AWS DocumentDB
Cloud & DevOps: AWS (ECS, EC2, Lambda, S3, RDS, Route 53, CloudFront, IAM), Transfer Family, Docker, Kubernetes, Jenkins, GitLab CI/CD
Security & Testing: OAuth2, JWT, JUnit, Pytest, Karate, Apache JMeter, Postman, Cucumber
Version Control & Tools: Git, GitLab, GitHub, Maven, IntelliJ, VS Code, STS
Methodologies: Agile/Scrum, CI/CD Automation, SOLID, OOP, Multithreading, TDD

Professional Experience

Bimbo Bakeries | Rhode Island

Full Stack Python & Java Developer (AI & Intelligent Automation) **March 2022 – Present**

- Developed AI-driven production intelligence modules using Java (Spring Boot) microservices that consumed Python-based ML inference APIs.
- Designed predictive maintenance solutions combining Java event processing with scikit-learn models trained in Python.
- Built a hybrid data pipeline that collected sensor data via Kafka, pre-processed it with Python, and pushed predictions to Java services for alert generation.
- Integrated TensorFlow and PyTorch models into Spring Boot applications to detect abnormal equipment behavior in near real time.
- Deployed AWS SageMaker endpoints and integrated them into Java REST APIs to automate production forecasting tasks.
- Created FastAPI services for Python-based inference, enabling seamless communication with Java orchestration layers.
- Implemented a rules engine in Java to interpret ML outputs and trigger process adjustments within manufacturing workflows.
- Designed a data ingestion framework in Java using AWS Kinesis and Lambda to capture and feed live plant metrics into ML models.
- Collaborated with data scientists to translate prototype notebooks into containerized, production-grade Python services callable from Java backends.

- Automated model retraining and deployment workflows using Jenkins pipelines that coordinated both Java and Python components.
- Built monitoring utilities in Java to track model response times, versioning, and inference accuracy within microservice clusters.
- Integrated AI-generated insights into React dashboards by connecting Java APIs to Python analytics endpoints.
- Created feature extraction modules in Python to prepare structured datasets for ML models used in production demand prediction.
- Utilized AWS Lambda and Java APIs for edge inference, delivering real-time decisions directly at manufacturing control points.
- Contributed to architectural blueprints that standardized cross-language AI service interaction between Java microservices and Python ML containers.

Charles Schwab | Birmingham, AL
Backend Developer (Python/Java) | Jan 2020 – Feb 2022

- Migrated backend systems from Java 8 to Java 11 and restructured Spring Boot applications using AOP, dependency injection, and modular service design.
- Designed REST APIs in Spring Boot to replace legacy SOAP interfaces, providing structured JSON endpoints for financial data exchange.
- Implemented CRUD operations in Java using Hibernate and JPA for secure and efficient data management in Oracle databases.
- Built AI data pre-processing modules in Python for normalization, cleanup, and feature extraction prior to model training.
- Developed Flask-based inference APIs in Python integrated with Java business services for real-time model scoring.
- Introduced early AI-assisted automation by embedding prediction endpoints within existing Spring Boot workflows.
- Utilized C++ for performance-critical components within backend processing modules that required high computation speed.
- Containerized Java, Node.js, and Python services using Docker and deployed across Kubernetes clusters with Helm for environment consistency.
- Created feature flag mechanisms in Java and Angular to toggle AI models dynamically during A/B testing phases.
- Managed database operations using Hibernate and optimized Oracle PL/SQL stored procedures to support financial transaction workflows.
- Built monitoring dashboards with Grafana and AWS CloudWatch to track API performance, inference latency, and application uptime.
- Developed secure backend logic using Node.js and TypeScript, maintaining consistent architecture and defensive programming practices.
- Automated deployment workflows through shell scripting on UNIX and managed JBOSS server configurations for hybrid environments.
- Collaborated with business analysts and product owners to design new AI-enabled modules supporting intelligent insights for trading systems.
- Participated in full software lifecycle phases including requirements analysis, UML design, development, testing, deployment, and documentation.

J.B. Hunt Transport Services | Arizona
Software Engineer **May 2019 – Dec 2019**

- Built microservices and REST APIs using Python (Flask, Django REST Framework) and Java (Spring Boot) for logistics applications.
- Created automated data processing pipelines using Pandas and NumPy for fleet analytics.
- Integrated AWS DocumentDB and PostgreSQL to support structured and unstructured logistics data storage.
- Implemented OAuth2 and JWT authentication for secure communication between logistics systems.
- Deployed containerized services on AWS ECS with Docker and Kubernetes for high availability.
- Developed React-based dashboards for shipment tracking and route optimization.
- Configured CI/CD pipelines using Jenkins and GitLab for reliable application delivery.
- Wrote Bash scripts for API monitoring, cron job automation, and server maintenance.
- Conducted unit and integration testing using PyTest and JUnit to maintain software reliability.
- Participated in Agile/Scrum sprints collaborating with developers, data scientists, and analysts.
- Performed performance profiling and optimized API response times across services.
- Authored API documentation using Swagger and OpenAPI for internal developer reference.
- Monitored distributed services using AWS CloudWatch and ELK stack for proactive issue resolution.
- Supported production troubleshooting through log analysis and service health diagnostics.
- Assisted in designing microservice interactions and data models to support logistics business functions.

Education

Master of Science in Information Technology | University of the Cumberlands | Williamsburg, KY
Bachelor of Science in Computer Science | East Central University | Ada, Oklahoma

Certificates:

AWS Certified Solutions Architect – Professional
Microsoft Certified: Azure AI Engineer Associate